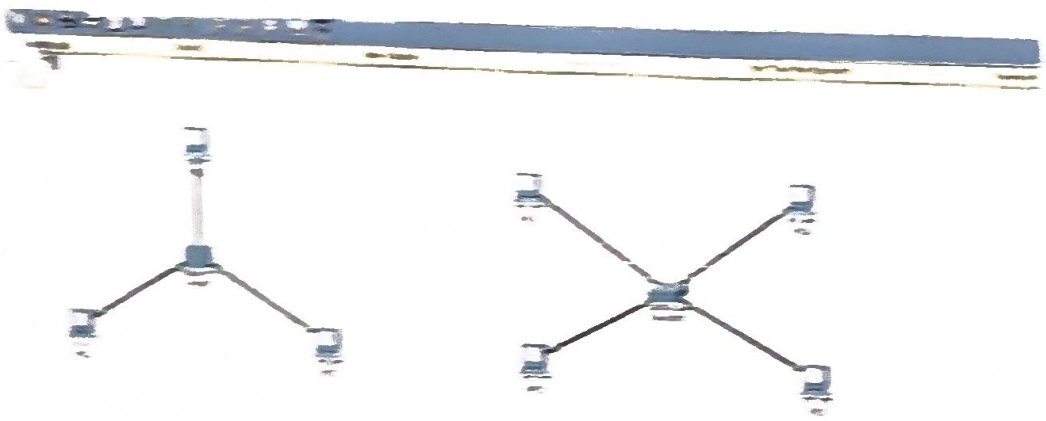


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23/07/26

Linux Networking Commands

1. IP

This IP command is one of the basic commands every administrator will need in daily work from setting up new system and assigning IPs to troubleshooting existing systems. The IP Command to sho information manipulation and tunnels.

IP <options> <object> <command>

Here are some common use cases for the IP command

(a) To show the ip addresses assigned to an interface of your server.

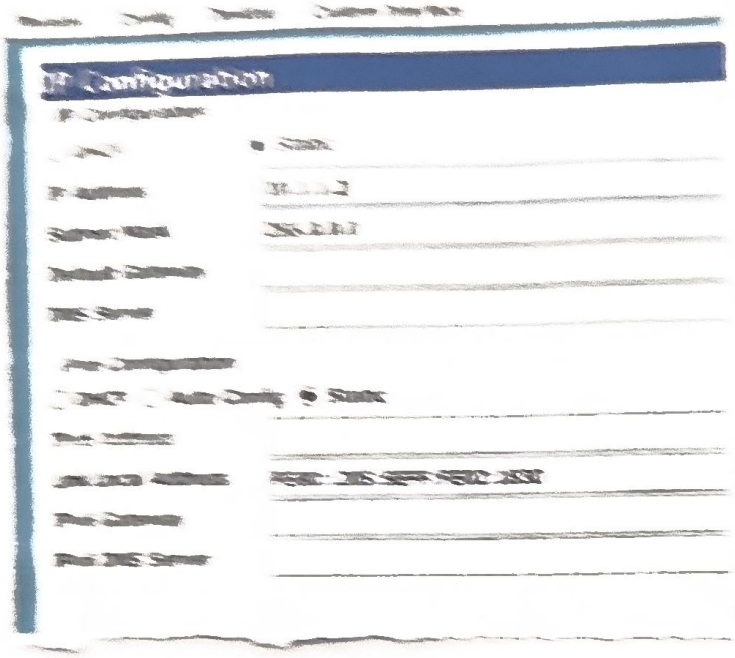
a. [root @ server ~] # ip address add 192.168.1.254/24 dev enps 03

(b) To assign an IP an interface for example enp03:

a. [root @ server ~]# ip address add 192.168.1.254/24 dev enp 03

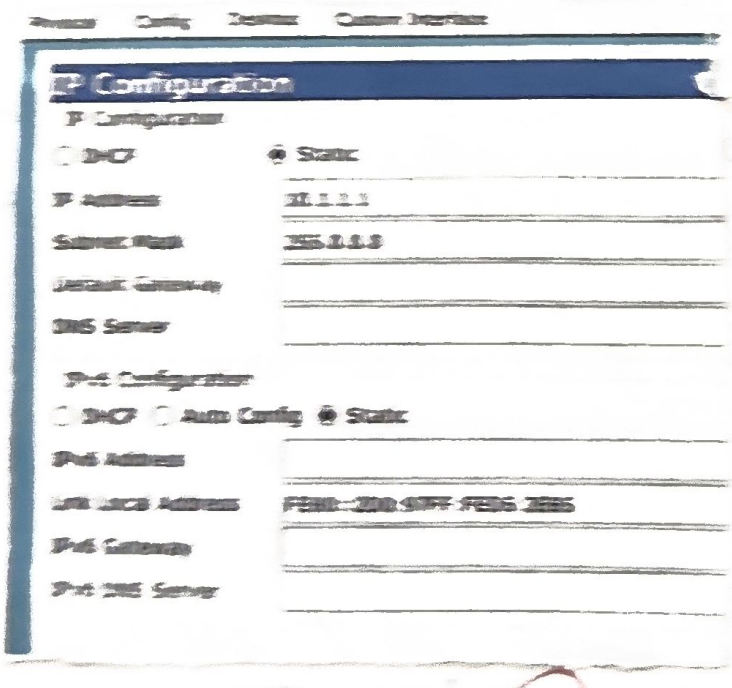
(c) To delete an IP on an Interface

a. [root @ server ~] # ip address del 192.168.1.254/24 dev enps03



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d. alter the status the interface by bringing the interface eth0 online:

[root@server ~]# ip link set eth0 down

e. Alter the status of the interface by bringing the interface eth0 offline:

[root@server ~]# ip link set eth0 promise on

f. alter the interface by enabling promiscuous mode of eth0

[root@server ~]# ip link set eth0 promisc on

g. Add a default route (for all addresses) via the local gateway 192.168.1.254 that can be reached on device eth0):

[root@server ~]# IP route add default via 192.168.1.254 dev eth0)

h. Display the route taken for ip 10.10.1.4:

[root@server ~]:# ip route get 10.10.1.4

ifconfig

The ifconfig command was/is a staple in many sysadmins tool belt for configuring and troubleshooting networks. It has now since been replaced by the ip command discussed above

mtr

MTR (mtr is traceroute) is a program with a command line interface that serves as a network diagnostic and troubleshooting tool this tool combines the functionality of the ping and traceroute commands. mtr provides a lot of statistics about each hop. such as response time and percentage

mtr <options> hostname/ip

lets look at some common use cases

a. The basic mtr command shows you the stats. including each hop (host name) with time & loss%.

[root@server ~]# mtr google.com

b. Show numeric IP address (if you use -g. you will get ip addresses (numbers) instead of hostnames

[root@server~]# mtr -b google.com

c. Show the numeric IP addresses and hostnames too

[root@server~]# mtr -b google.com

4 tcpdump

the tcpdump command is designed for capturing and displaying packets

you can install tcpdump with the command below

[root@server~]# dnf install -y tcpdump

Before starting any capture you need to know which interfaces tcpdump can use you will need to use sudo or have root access in the case

[root@server~]# tcpdump -D

If you wand to capture traffic eth0. you can initiate tat with tcpdump -i eth0 sample

output:

[root@server~]# tcpdump -i eth0

5. Ping

Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet control message protocol (icmp) echo Request messages. The reciept of corresponding Echo reply messages troubleshoot connectivity reachibility and